METRIC

MIL–PRF–51527A(EA)
13 March 2000

SUPERSEDING
MIL–F–51527(EA)
22 May 1990,
MIL–F–51525
15 July 1996

PERFORMANCE SPECIFICATION

FILTER SET, GAS-PARTICULATE, 340 CMH (200 CFM)

Reactivated for New Design after 13 March 2000

This specification is approved for use by the U.S. Army Soldier and Biological Chemical Command (SBCCOM), Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers 2 types of gas-particulate filter sets rated at 340 cubic meters per hour (cmh) (200 cubic feet per minute (cfm)) for use in collective protection configurations (see 6.1):

- Type I - Filter set, gas-particulate, 340 cmh (200 cfm) for Advanced Integrated Collective Protection Systems (AICPS).
- Type II - Filter set, gas-particulate, 340 cmh (200 cfm) for land based Collective Protection Equipment (MCPE), for shipboard Collective Protection Systems (CPS), and for the fan filter assembly of the transportable CPS.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Technical Director, U.S. Army Edgewood Chemical Biological Center, ATTN: AMSSB–REN–SS, Aberdeen Proving Ground, MD 21010–5424 by using the self–addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4240

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Each filter set consists of a separate gas and particulate filter. This specification should be used in conjunction with MIL-PRF-51526 on particulate filters for the purchase of gas-particulate filter sets. No separate specification for gas filters is available.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to insure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL−PRF-51526	–	Filter, Particulate 340 cmh (200 cfm)
MIL-S-901	–	Shock Test, High-Impact Shipboard Machinery, Equipment, and Systems, Requirements for

STANDARDS

DEPARTMENT OF DEFENSE

MIL−STD−167−1	–	Mechanical Vibrations of Shipboard Equipment
MIL−STD−282	–	Military Standard, Filter Units, Protective Clothing, Gas−Mask Components and Related Products: Performance Test Methods
MIL−STD−810	–	Environmental Test Methods and Engineering Guidelines

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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U.S. ARMY EDGEWOOD CHEMICAL BIOLOGICAL CENTER

DRAWINGS

5–19–6120	–	Housing, Gas–Particulate Filter, 1 Filter
5–19–6121	–	Housing, Gas–Particulate Filter, 2 Filter
5–19–12983	–	Housing, NBC Filter, 200 CFM

PURCHASE DESCRIPTIONS

EA–DTL–1704	–	Carbon, Activated, Impregnated, Copper-Silver-Zinc-Molybdenum-Triethylenediamine (ASZM-TEDA)

(Copies are available from Technical Director, U.S. Army Edgewood Chemical Biologcal Center, ATTN: AMSSB–REN–SE, Aberdeen Proving Ground, MD 21010–5424.)

NAVAL SEA COMMAND

DRAWINGS

53711–6263500	–	Filter System, CPS
53711–6573697	–	Filter System, Navy Shipboard Selected Area Collective Protection System (SACPS)

SS200–AG–MMM–010	–	Navy Shipboard Collective Protection System(CPS) Technical Manual, CBR Filter System Operation and Maintenance
SS200–AL–MMM–010	–	Technical Manual for Navy Shipboard Selected Area Collective Protection System (SACPS) System Description, Operation, and Maintenance

(Copies are available from Naval Surface Warfare Center Dahlgren Division, 17320 Dahlgren Road, Code G52, Dahlgren, VA 22448–5100.)

CODE OF FEDERAL REGULATIONS

40 CFR Part 261 – Identification and Listing of Hazardous Waste

(Copies are available from the U.S. Government Printing Office, Washington, DC 20402)

2.3 Non–Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless other-

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wise specified, the issues of documents not listed in the DoDISS are the issue of the documents cited in the solicitation (see 6.2).

AIR–CONDITIONING AND REFRIGERATION INSTITUTE

ARI 700 – Specifications for Fluorocarbon and Other Refrigerants

(Application for copies should be addressed to Global Engineering Documents, 15 Inverness Way East, Englewood, CO 80112–5704.) (http://global.ihs.com)

AMERICAN CONFERENCE OF GOVERNMENTAL INDUSTRIAL HYGIENISTS

Publication 0099 – Threshold Limit Values and Biological Exposure Indices

(Application for copy should be addressed to 1330 Kemper Meadow Dr., Suite 600, Cincinnati, OH 45240.) (http://www.acgih.org/)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B209 – Standard Specification for Aluminum and Aluminum Alloy Sheet and Plate

ASTM D1056 – Standard Specification for Flexible Cellular Materials – Sponge or Expanded Rubber

ASTM D2867 – Moisture in Activated Carbon

(Application for copies should be addressed to ASTM, 100 Barr Harbor Drive, West Conshohocken, PA 19428–2959.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. The filter set shall be constructed so that it conforms to the requirements of this specification. The materials of construction shall not be capable of having any adverse effects on human health during normal use. The contractor shall select materials which pose no potential inhalation hazard to the user, and whose normal use will not exceed the 8–hour time weighted average threshold limit values published by the American Conference of Governmental Industrial Hygienists (ACGIH) (see 6.2). At the time of disposal, unused and uncontaminated filters shall not be a Resource Conservation Recovery Act hazardous waste according to

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characteristics cited in 40 CFR 261.21−261.24 and shall not be listed as specific hazardous waste chemicals in 40 CFR 261.33(e) and (f).

3.1.1 Adsorbent media, gas filter. The adsorbent media shall fully meet all of the requirements in the latest version of EA−DTL−1704 for ASZM−TEDA carbon.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspections in accordance with 4.2. The particulate filter must meet all requirements specified in MIL−PRF−51526.

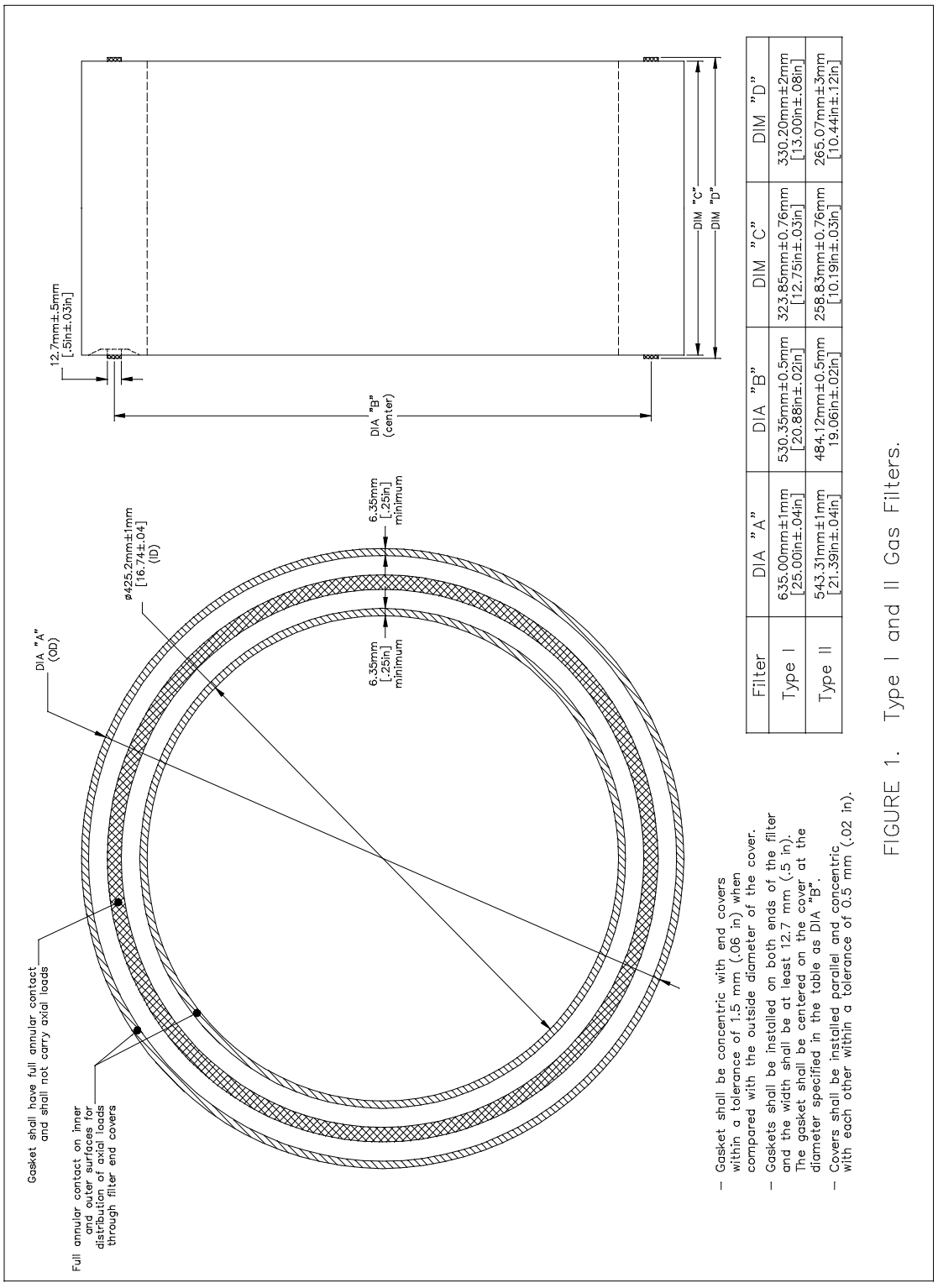
3.3 Interface requirements.

3.3.1 Compatibility with filter housing. Each filter in the set shall be cylindrical in form and configured so that air shall flow through the filter radially outward for compatibility with its intended filter housing.

3.3.1.1 Type I filters. Each filter set shall contain one gas and one particulate filter that are separable and not permanently attached to each other. The filter set shall be of a size that fits inside the intended filter housing (Drawing 5−19−12983). Each filter shall fit easily into the housing and be removed with no damage to itself or the surrounding components. The particulate filter shall meet all interface requirements for a Type I filter stated in MIL−PRF−51526. The gas filter must meet the interface dimensions in Figure 1. Each end of the filter or end cover shall contain one groove for gasket location. The gasket shall be grade number 2C1 of ASTM D1056, seamless, have a thickness of 0.787 cm (0.31 in) and be made of a material which has fuel and chemical agent permeation rates no higher than those of neoprene. The gasket shall be permanently attached to the filter with an adhesive.

3.3.1.2 Type II filters. Each filter set shall contain one gas and one particulate filter that are separable and not permanently attached to each other. The filter set shall be of a size that fits inside the intended filter housings (Drawings 5−19−6120, 5−19−6121, and 53711−6263500). Each filter shall fit easily into the housing and be removed with no damage to itself or the surrounding components. The particulate filter shall meet all interface requirements for a Type II filter stated in MIL−PRF−51526. The gas filter must meet the interface dimensions in Figure 1. Each end of the filter or end cover shall contain one groove for gasket location. The gasket shall be grade number 2C1 of ASTM D1056, seamless, have a thickness of 0.787 cm (0.31 in) and be made of a material which has fuel and chemical agent permeation rates no higher than those of neoprene. The gasket shall permanently attached to the filter with an adhesive.

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Filter	DIA "A"	DIA "B"	DIM "C"	DIM "D"
Type I	635.00mm±1mm [25.00in±.04in]	530.35mm±0.5mm [20.88in±.02in]	323.85mm±0.76mm [12.75in±.03in]	330.20mm±2mm [13.00in±.08in]
Type II	543.31mm±1mm [21.39in±.04in]	484.12mm±0.5mm 19.06in±.02in]	258.83mm±0.76mm [10.19in±.03in]	265.07mm±3mm [10.44in±.12in]

- Gasket shall be concentric with end covers within a tolerance of 1.5 mm (.06 in) when compared with the outside diameter of the cover.
- Gaskets shall be installed on both ends of the filter and the width shall be at least 12.7 mm (.5 in). The gasket shall be centered on the cover at the diameter specified in the table as DIA "B".
- Covers shall be installed parallel and concentric, with each other within a tolerance of 0.5 mm (.02 in).

FIGURE 1. Type I and II Gas Filters.

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3.3.2 Interchangeability. Each gas and particulate filter and any associated gasket shall be individually interchangeable (replaceable) by one of similar form, fit and function without modification of the filter housing.

3.3.3 Weight. The weight of a Type I filter set shall be no greater than 45.4 kilograms (100 pounds) while unpackaged. The weight of a Type II filter set shall be no greater than 22.7 kilograms (50 pounds) while unpackaged and no greater than 31.8 kilograms (70 pounds) while packaged.

3.4 Operating requirements.

3.4.1 Airflow resistance. The gas filter shall have a rated flow of 340 cmh (200 cfm) at standard conditions (21°C and 1 atm (70°F and 1013 mbar)). The airflow resistance shall not exceed the airflow resistance (pressure drop) of 14.3 centimeters (5.6 inches) water gage (w.g.) for the Type I gas filter and 11.4 centimeters (4.5 inches w.g.) for the Type II gas filter when airflow is 340 cmh (200 cfm) at standard conditions. See MIL–PRF–51526 for airflow resistance requirements for particulate filter.

3.4.2 Leakage. The gas filter of Type I and II filter sets shall not leak when a concentration of 1000 parts per million (ppm) of R–134a (1,1,1,2) tetrafluoroethane refrigerant is introduced at the filter inlet. A filter leak is defined as the presence of 1 ppm or more of R–134a in the effluent air within 2 minutes after the introduction of the refrigerant at the inlet.

3.4.3 Gas life, Cyanogen Chloride (CK). Each sample adsorbent specimen obtained from the filling process of the gas filter shall have a CK life of no less than 95% of that specified in EA–DTL–1704.

3.4.4 Gas life, Dimethylmethylphosphonate (DMMP). The DMMP life of the gas filter shall be at least 200 minutes if the filter is Type I and at least 60 minutes if the filter is Type II when the influent DMMP concentration is 5000 mg/m^3 at rated flow.

3.5 Environmental requirements.

3.5.1 Resistance to hot/cold temperature. Each filter set shall show no ignition or structural damage as a result of temperature conditioning between –51 to 71°C (–60 to 160°F) and be capable of meeting operating requirements in 3.4.1 and 3.4.2 thereafter.

3.5.2 Structural integrity.

3.5.2.1 Resistance to structural deformation. The gas filter and particulate filter shall be capable of withstanding repeated applications of a quasi–static mechanical load of 12900 newtons (2900 pounds) in the axial direction. Each gasket shall not become dislodged from the filter by the process of loading a filter into a housing and removing it.

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3.5.2.2 Resistance to shock. After shock treatment, the gas filter shall meet the DMMP life requirement in 3.4.4 and the particulate filter shall meet the filtration efficiency requirement in MIL–PRF–51526.

3.5.2.3 Resistance to vibration. After vibration treatment, the gas filter shall meet the DMMP life requirement in 3.4.4 and the particulate filter shall meet the filtration efficiency requirement in MIL–PRF–51526.

3.5.2.4 Resistance to rough handling. After rough handling consisting of two hundred 19–millimeter drops in 15 minutes, the gas filter shall meet the DMMP life requirement in 3.4.4 and the particulate filter shall meet the filtration efficiency requirement in MIL–PRF–51526.

3.5.3 Resistance to fungus and mildew. The materials of construction shall not support fungus and mildew growth beyond trace levels.

3.5.4 Resistance to corrosion. Any metallic construction material shall resist salt fog corrosion to an extent equal or greater than anodized aluminum when the aluminum alloy number is in the 3000 to 5000 series of ASTM B209.

3.5.5 Resistance to overpressure. The particulate filter shall meet the overpressure requirements in MIL–PRF–51526.

3.5.6 Dust capacity. The particulate filter shall meet the dust capacity requirements in MIL–PRF–51526.

3.5.7 Flammability. The particulate filter shall meet the flammability requirements in MIL–PRF–51526.

3.6 Ownership and support.

3.6.1 Moisture content. To enhance storage of the gas filter, the humidity conditions during manufacturing, materials handling, and packaging shall be such that the moisture content of adsorbent in the packaged gas filter shall not exceed 3.0 percent by weight.

3.6.2 Identification markings. Each filter in the set shall contain identification markings. Markings shall be legible using contrasting ink and ½ inch high bold font. The ink shall retain its color and legibility after being exposed to water for 4 hours. Markings shall include the name of the item, lot number, part number and any other information required by the acquisition document (see 6.2). The name marking on the gas filter shall be “FILTER, GAS, 340 CMH (200 CFM)” and the name marking on the particulate filter shall be “FILTER, PARTICULATE, 340 CMH (200 CFM).”

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3.6.3 Workmanship

3.6.3.1 Safety. Filters shall be free from burrs and sharp edges that might injure personnel, damage filter media, or damage Mission Oriented Protective Posture (MOPP) IV gloves during handling.

3.6.3.2 Finish. The filters shall be free from foreign matter (dirt, oil, or viscous materials).

3.6.3.3 Appearance. The filter units shall be free from cracked, bent, or dented metal sections, and abraded gaskets.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2)
(b) Conformance inspection (see 4.3)

4.2 First article inspection.

4.2.1 Sample. The first article (FA) sample shall consist of 20 filter sets that are of the type specified in the contract. The FA sample shall be produced using the same methods, materials, equipment, and processes as will be used during regular production.

4.2.2 Inspections to be performed. Unless otherwise specified by the terms of the contract, the sample first article items shall be subjected to all of the examinations and tests specified in this specification. Verification of requirements shall be by visual inspection, commercial inspection equipment, demonstration, test data, or by Certificate of Compliance (COC) with supporting evidence or analysis. The Government reserves the right to accept or reject each COC with supporting evidence or analysis.

4.2.2.1 For examination. Sample filters shall first be examined for interface and other characteristics in Table I and Figure 1 that may be verified by visual inspection, commercial inspection equipment or simple demonstrations.

4.2.2.2 For test.

4.2.2.2.1 Carbon testing.

4.2.2.2.1.1 Samples for moisture testing. A sample of adsorbent shall be taken during the filling of the 1st, 10th, and 20th filter. Each carbon sample taken during the filling of those 3 filters shall weigh at least 100 grams (0.22 pounds). Each sample shall be placed in an open container whose diameter is such that the depth of the carbon will be 3.81± 1.27 cm (1.5± 0.5) inches deep. Those samples of carbon shall be exposed to the same conditions of temperature and humidity as

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their associated filters and be used for subsequent moisture testing. Concurrent with or immediately following the packaging of the filters corresponding to each carbon sample, each of 3 carbon samples shall be tested for moisture content in accordance with 4.3.6.10. If any carbon sample fails to meet the moisture requirement in 3.6.1, the first article lot of filters represented by the samples shall be rejected.

4.2.2.2.1.2 Samples for CK gas life testing. A sample of adsorbent shall be taken during the filling of each of 8 first article filters, placed in an open container, and exposed to the same conditions of temperature and humidity as its associated filter. Each adsorbent sample taken during the filling of those filters shall weigh at least 100 grams (0.22 pounds). The 8 samples shall then be sealed in their containers and sent to the Government for subsequent CK life testing. The Government will determine the minimum CK gas life in accordance with 4.3.6.4 and EA-DTL-1704. If any sample fails to meet the requirements for CK life in EA-DTL-1704, the first article lot represented shall be rejected.

4.2.2.2.2 Filter testing. First article samples shall be tested in accordance with Table I and Figure 2. Following interface checks, airflow resistance, leakage testing of sample filters, moisture content testing of adsorbent material, and airflow resistance and filtration efficiency testing of particulate filters, sample filters shall be packaged using the same methods, materials, equipment, and processes as will be used during regular production. Packaged filter sets shall then be forwarded to the Government for environmental tests as depicted in Figure 2.

4.2.3 Acceptance criteria. If any first article sample item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

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TABLE I. First Article Tests and Inspection

Test Description	Requirement Paragraph	Verification Paragraph
Materials	3.1	Certificate of Compliance
Interface requirements	3.3	
Compatibility with filter housing	3.3.1	4.3.6.1
Interchangeability	3.3.2	4.3.6.1
Weight	3.3.3	4.3.6.1
Operating requirements	3.4	
Airflow resistance	3.4.1	4.3.6.2
Leakage	3.4.2	4.3.6.3
Gas life, CK	3.4.3	4.3.6.4
Gas life, DMMP	3.4.4	4.3.6.5
Environmental requirements	3.5	
Resistance to hot/cold temperature	3.5.1	4.3.6.6
Structural integrity	3.5.2	4.3.6.7
Resistance to structural deformation	3.5.2.1	4.3.6.7.1
Resistance to shock	3.5.2.2	4.3.6.7.2
Resistance to vibration	3.5.2.3	4.3.6.7.3
Resistance to rough handling	3.5.2.4	4.3.6.7.4
Resistance to fungus and mildew	3.5.3	4.3.6.8
Resistance to corrosion	3.5.4	4.3.6.9
Resistance to overpressure*	3.5.5	MIL-PRF-51526
Dust capacity*	3.5.6	MIL-PRF-51526
Flammability*	3.5.7	MIL-PRF-51526
Ownership and support	3.6	
Moisture content	3.6.1	4.3.6.10
Identification markings	3.6.2	VI
Workmanship	3.6.3	
Safety	3.6.3.1	VI & demo
Finish	3.6.3.2	VI
Appearance	3.6.3.3	VI

* Applicable to particulate filter only

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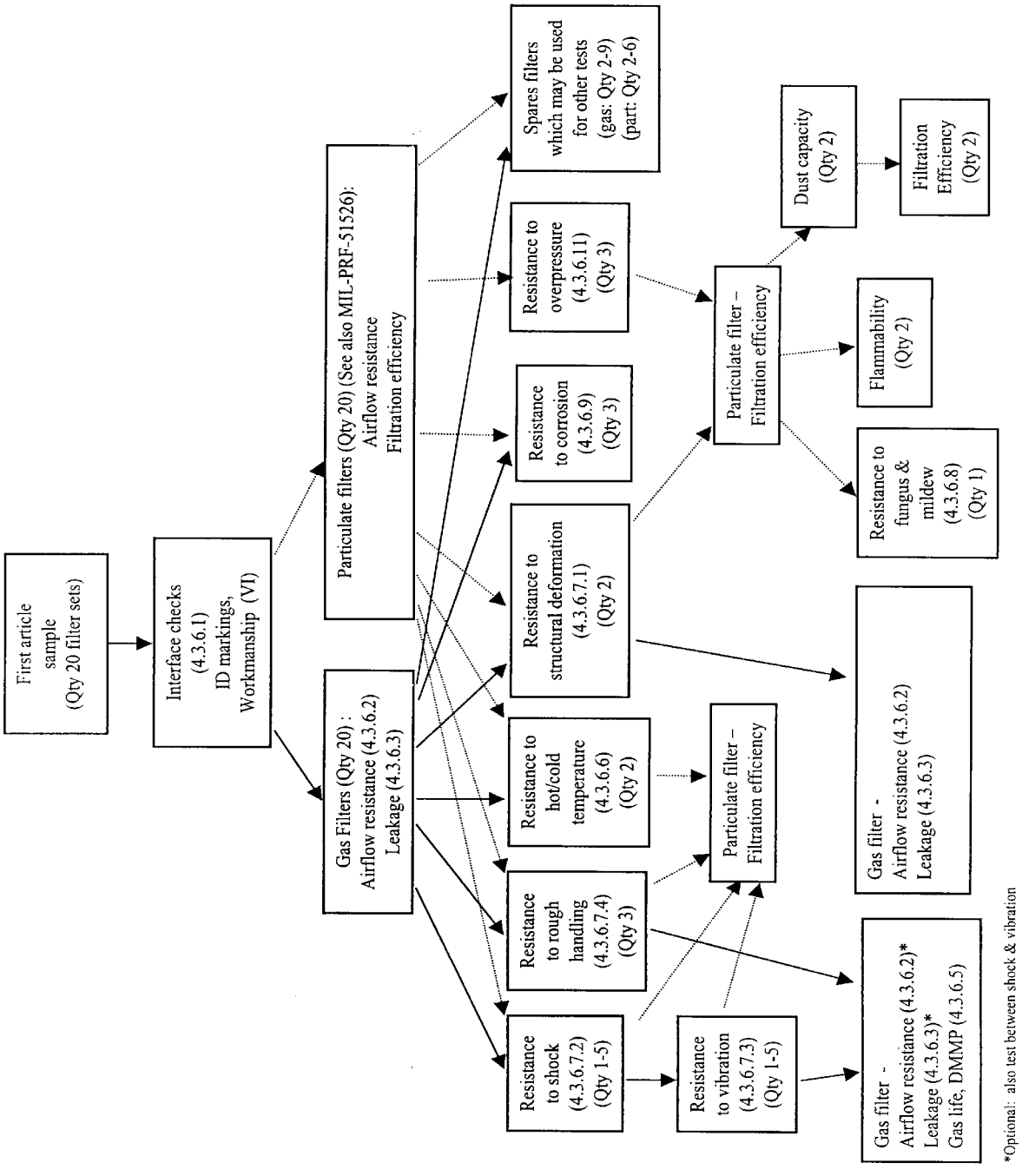


FIGURE 2. First Article Inspection and Test Sequence, Type I and II

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4.3 Conformance inspection.

4.3.1 Lotting. A lot shall consist of a series of filter sets whose separate gas and particulate filters are each produced by one manufacturer, at one plant, from the same materials, under the same manufacturing and process conditions and without a break in production of more than 10 calendar days. The gas and particulate filters in a given lot of filter sets need not be produced from the same manufacturer. Any one lot of gas filters shall contain no more than one lot of adsorbent media. Each lot shall be identified by an alphanumeric lot number. The lot number shall include a manufacturer's identification symbol consisting of 3 alpha characters, a numeric code identifying the year of production, a code or abbreviation that signifies the month of production, and an interfix–serial number. The interfix–serial number shall change if there is a change in the design, manufacturing process, materials, suppliers, production run, or if a new contract is used.

4.3.2 Sampling. Inspection and test of random samples shall be conducted in accordance with the classification of characteristics in 4.3.5, and when specified in Table II.

4.3.3 Inspection procedure. Every item in the lot shall be inspected for critical characteristics. Sample filters shall be examined and tested in accordance with the classification of characteristics in 4.3.5. If a filter is found that does not conform to any characteristic inspected 100%, such as leakage and pressure drop requirements, the non–conforming filter shall be rejected and removed from the lot. For characteristics other than leakage and pressure drop, failure of any sample filter to conform to any characteristic in the classification of characteristics based on the sampling and acceptance criteria specified therein shall be cause for rejection of the lot represented (see footnote, Table II).

4.3.3.1 Carbon testing.

4.3.3.1.1 Samples for moisture testing. A sample of adsorbent shall be taken during the filling of the first and last filter manufactured each day. Each carbon sample taken during the filling of those 2 filters shall weigh at least 100 grams. Each sample shall be placed in an open container whose diameter is such that the depth of the carbon will be 3.81± 1.27 cm (1.5± 0.5) inches deep. Those samples of carbon shall be exposed to the same conditions of temperature and humidity as their associated filters and be used for subsequent moisture testing. Concurrent with or immediately following the packaging of the filters corresponding to each carbon sample, each carbon sample shall be tested for moisture content in accordance with 4.3.6.10. If any carbon samples fails to meet the moisture requirement in 3.6.1, the production run of filters fabricated that day shall be rejected.

4.3.3.1.2 Samples for CK life testing. For each lot, 8 samples of adsorbent, each sample weighing at least 100 grams, shall be taken at random times during the filling of 8 respective filters and placed in an open container. Those samples shall be exposed to the same conditions of temperature and humidity as their associated filters. The 8 samples shall then be sealed in their

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containers and sent to the Government for subsequent CK life testing. The Government will determine the minimum CK life in accordance with 4.3.6.4 (EA–DTL–1704).

4.3.3.1.3 Filter testing. Each gas filter shall be tested in accordance with tests listed in Classification of Characteristics table in 4.3.5. Any filter samples forwarded to the Government for testing shall be packaged such they will be protected from moisture in the air. Each gas filter shall be tested for airflow resistance (4.3.6.2) and leakage (4.3.6.3). Following leakage and airflow resistance testing, sample filters shall be packaged using the same methods, materials, equipment, and processes as will be used during regular production. After sample gas filters are tested for moisture content in accordance with 4.3.6.10, initial airflow resistance (4.3.6.2), leakage (4.3.6.3), and after sample particulate filters are tested for filtration efficiency in accordance with MIL–PRF–51526, the packaged filter sets shall then be forwarded to the Government for additional tests as depicted in Figure 2.

4.3.4 Inspection characteristics. Critical characteristics are those whose nonconformance to specified requirements is likely to result in hazardous or unsafe conditions for individuals using, maintaining, or depending upon the product, characteristics whose nonconformance to specified requirements is likely to prevent performance of the tactical function of a major end item. Major characteristics are those whose nonconformance to specified requirements is likely to result in failure or to reduce materially the usability of the item for its intended purpose. Minor characteristics are those whose nonconformance to specified requirements is not likely to reduce materially the operation or usability of the item for its intended purpose.

4.3.5 Classification of characteristics. Conformance examinations and tests shall be as specified in Table III. When specified herein, accept on “0” and reject on “1” attributes sampling inspection shall be performed on the designated characteristics using the stated levels in Table II for selection of sample sizes.

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TABLE II. Sampling

Lot size	Inspection levels and sample sizes										
	I	II	III	IV	V	VI	VII	VIII	IX	X	XI
2 to 8	*	*	*	*	*	*	*	*	5	3	2
9 to 15	*	*	*	*	*	*	13	8	5	3	2
16 to 25	*	*	*	*	*	20	13	8	5	3	3
26 to 50	*	*	*	*	32	20	13	8	5	5	5
51 to 90	*	*	*	50	32	20	13	8	7	6	5
91 to 150	*	*	125	50	32	20	13	12	11	7	6
151 to 280	*	*	125	50	32	20	20	19	13	10	7
281 to 500	*	315	125	50	48	47	29	21	16	11	9
501 to 1200	*	315	125	75	73	47	34	27	19	15	11
1201 to 3200	1250	315	125	116	73	53	42	35	23	18	13
3201 to 10000	1250	315	192	116	86	68	50	38	29	22	15
10001 to 35000	1250	315	294	135	108	77	60	46	35	29	15
35001 to 150000	1250	490	294	170	123	96	74	56	40	29	15
150001 to 500000	1250	715	345	200	156	119	90	64	40	29	15
500001 and over	1250	715	435	244	189	143	102	64	40	29	15

*Indicates one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection.
Accept the lot represented on zero nonconforming characteristics and reject the lot represented on one or more nonconforming characteristics for all inspection levels.

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TABLE III. Classification of Characteristics

PARAGRAPH 4.3.5	TITLE Filter, gas*		Sheet 1 of 1	DRWG NO.
CATEGORY	CHARACTERISTIC	SAMPLING	REQUIRE-MENT PARAGRAPH	INSPECTION METHOD
Critical	Leakage	100%	3.4.2	4.3.6.3
Major				
101	Airflow resistance	100%	3.4.1	4.3.6.2
102	Gas life, CK**	See 4.3.6.4	3.4.3	4.3.6.4
103	Gas life, DMMP** after rough handling	Table II, Level XI	3.4.4	4.3.6.5
104	Moisture content	See 4.3.6.10	3.6.1	4.3.6.10
106	Safety	Table II, Level IX	3.6.3.1	VI & D
Minor				
201	Identification markings	Table II, Level IX	3.6.2	VI
202	Finish	Table II, Level IX	3.6.3.2	VI
203	Appearance	Table II, Level IX	3.6.3.3	VI

NOTES:
CE – Commercial inspection equipment
COC – Certificate of Compliance
VI – Visual inspection
D – Demonstration

*For required characteristics of the particulate filter, see MIL-PRF-51526
**Failure may be considered a critical defect

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4.3.6 Test.

4.3.6.1 Interface checks. Determine whether each sample filter complies with interface requirements in 3.3 by inspection of dimensions and commercial inspection equipment. Perform installation/removal demonstrations of sample filter sets with the filter housing if filter housing is Government furnished (see 6.2).

4.3.6.2 Airflow resistance (pressure drop). The airflow resistance shall be determined at the rated airflow of 340 cmh (200 cfm). The measured pressure drop across the particulate filter, when corrected to standard conditions of 21°C (70°F) and 1 atm (1013 mbar), shall be no greater than that specified in 3.4.1. The up–stream and down–stream static pressure measuring tubes shall be as close as possible to the filter and shall not be on a section of duct that has a changing cross sectional area. Test for airflow resistance as follows:

(a) Connect the filter to a source of forced air.
(b) Set the radially outward flow of air through the filter to 340 cmh.
(c) Record the barometric pressure.
(d) Measure and record the air stream temperature. This temperature should be no higher than 52°C (125°F).
(e) Determine and record the difference between up-stream and downstream static pressure.

If the recorded air stream temperature and barometric pressure is different than 21°C (70°F) and 1 atm, calculate and record the air flow resistance (DP (cal)) using the equation below to correct the measured airflow resistance to airflow resistance at standard conditions:

$$DP\ (cal) = DP\ (measured) \times P\ (test) \times 86.21\ ,\ [(492 + 1.8T)^{1.768}]$$

where:

DP (cal) – air flow resistance corrected to standard conditions in millimeters of water gage (mm wg).
P (test) – barometric pressure at time of test in millimeters of mercury (mm Hg).
DP (measured) – air flow resistance from test measurement in millimeters of water.
T – temperature of air stream flowing through the filter in degrees C.

Note: Correction for nonstandard conditions is not required when it is clear that the passage or failure of the filter is not in doubt.

4.3.6.3 Leakage. Rigidly position the sample gas filter in the test fixture. Orient the filter in the fixture with the cylindrical axis in the horizontal direction and end faces in the vertical direction. Use R–134a refrigerant that conforms to the characteristics of ARI STANDARD 700. Connect the filter to a source of forced air and place a mixing chamber at the influent side. Regulate the flow of air from the blower to 1/5th the rated airflow of the filter (68 cmh). Introduce R–134a into the intake of the blower; monitor and maintain a concentration of 1000

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ppm by volume of R−134a in airflow of 68 cmh on the influent side of the filter at 24± 5°C (75 ± 9°F) and relative humidity of no higher than 50 percent. Leakage shall be determined using a suitable leak detector (see 6.4) for sampling and detecting R−134a at the effluent side of the filter. Monitor the effluent airstream for at least 2 minutes. Determine whether the effluent concentration exceeds 1 ppm within 2 minutes after the introduction of the refrigerant at the inlet of the filter. Purge the filter by passing clean air in the reverse direction through the filter at rated flow for 3 minutes. Keep the exposure of the filter to air to a minimum.

4.3.6.4 Gas life, CK. Perform CK life testing on each carbon sample placed in tubes in accordance with procedures, conditions and parameters in EA−DTL−1704.

4.3.6.5 Gas life, DMMP. DMMP gas life tests will be performed on filters that have previously undergone tests for airflow resistance, leakage, shock and vibration.

a. If sample item is a Type I or II gas filter, install gas filter in test tunnel in its service orientation with the identification label on the upper half of the filter. Adjust airflow to rated flow. Determine the DMMP gas life (the total time from the start of the DMMP challenge until the breakthrough concentration is reached) of the filter at the following test conditions:

   Challenge concentration of DMMP: 5000± 400 mg/m^3
   Breakthrough concentration of DMMP: 0.04 mg/m^3
   Relative humidity: less than or equal to 50 %
   Airstream and filter temperature: 47± 5 °C (116± 9 °F)

b. The DMMP challenge concentration in the air shall be monitored to ensure uniform mixing of the DMMP vapor with the air once it reaches the test item.

c. While the challenge concentration is permitted to vary within the stated range during the course of the test, the breakthrough time measured must be normalized to the 5000 mg/m^3 concentration by the following calculation:

$$\text{Corrected breakthrough time} = \text{Measured breakthrough time} * \frac{\text{Average concentration during test}}{5000\ \text{mg/m}^3}$$

4.3.6.6 Resistance to hot/cold temperature. Condition each filter set at ambient, (21± 2°C) (70± 3.5°F); high, (71± 2°C) (160± 3.5°F); and low, (−51± 2°C) (−60± 3.5°F) temperature for 4 hours at each temperature setting. After removal from the environmental chamber, inspect filter set for temperature related damage such as ignition, charring and discoloration. Then test each gas filter for airflow resistance and leakage in accordance with 4.3.6.2 and 4.3.6.3, respectively, and test each particulate filter for filtration efficiency in accordance with MIL−PRF−51526.

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4.3.6.7 Structural integrity.

4.3.6.7.1 Resistance to structural deformation. The particulate and gas filters (Type I and II) shall be subjected to a quasi–static mechanical load as specified in 3.5.2.1 in the axial direction for 3 cycles of one minute per cycle. Then test gas filters for airflow resistance and leakage in accordance with 4.3.6.2 and 4.3.6.3, respectively, and test particulate filters for filtration efficiency in accordance with MIL–PRF–51526.

4.3.6.7.2 Resistance to shock.

4.3.6.7.2.1 Type I filters. The particulate and gas filter shall be subjected to simulated bench handling in accordance with Method 516.4, Procedure VI of MIL-STD-810.

4.3.6.7.2.2 Type II filters. Shock testing shall be performed on gas-particulate filter sets when fully loaded in either the U.S. Navy CPS Filter System (53711-6263500), or the U.S. Navy SACPS Filter System (53711-6573697). The CPS system has a flow capacity of 1,020 cmh (600 cfm), and the SACPS system has a flow capacity ranging from 340 cmh (200 cfm) to 1700 cmh (1000 cfm). Filters shall be installed per the operation and maintenance manual for the tested system: SS200-AG-MMM-010 for the CPS system and SS200-AL-MMM-010 for the SACPS system. The filter housing shall be filled to capacity. The filter system (housing with particulate and gas filters) shall be shock tested per MIL-S-901 using an approved mounting fixture.

4.3.6.7.3 Resistance to vibration.

4.3.6.7.3.1 Type I filters. Vibration testing shall be performed on the gas and particulate filter MIL-STD-810, Method 514.4, category 8 using the levels and frequencies in Table 514.4-AII. For operational tests that follow shock and vibration, see Figure 2.

4.3.6.7.3.2 Type II filters. Vibration testing shall be performed on gas-particulate filter sets when loaded in either the U.S. Navy CPS Filter System (53711-6263500), or the U.S. Navy SACPS Filter System (53711-6573697). The CPS system has a flow capacity of 1,020 cmh (600 cfm), and the SACPS system has a flow capacity ranging from 340 cmh (200 cfm) to 1700 cmh (1000 cfm). Filters shall be installed per the operation and maintenance manual for the tested system: SS200-AG-MMM-010) for the CPS system and SS200-AL-MMM-010 for the SACPS system. The filter housing shall be filled to capacity. The filter system (housing with filter sets) shall be vibration tested per MIL-STD-167-1 using an approved mounting fixture. For operational tests that follow shock and vibration, see Figure 2.

4.3.6.7.4 Resistance to rough handling. Perform rough handling testing on separate gas and particulate filters that have not previously undergone shock and vibration testing. Clamp each filter to a steel plate such that the cylindrical axis of the filter will be parallel to the direction of movement. Perform rough handling consisting of 200 19–millimeter drops per minute for 15 minutes in accordance with procedures in Method 105.11 of MIL–STD–282. For operational tests that follow shock and vibration, see Figure 2.

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4.3.6.8 Resistance to fungus and mildew. The manufacturer shall provide objective evidence and certify that the materials of construction will not support fungus growth beyond trace levels for at least 28 days. Objective evidence includes historical data on similar construction materials, test data on filter construction materials, or test data on the entire filter system. If testing is conducted, the fungus resistance test shall be performed in accordance with Method 508.4 of MIL−STD−810 and the fungus exposure period shall be 28 days.

4.3.6.9 Resistance to corrosion. The manufacturer shall provide objective evidence and certify that any metallic construction materials will resist corrosion to an extent equal or greater than anodized aluminum or aluminum alloys in the 3000 to 5000 series of ASTM B209. Objective evidence includes test data on entire filter set. If objective evidence does not exist or is disapproved by the Government, perform the following tests: conduct salt fog testing in accordance with Method 509.3 of MIL−STD−810 except the total duration shall be 20 days corresponding to five 96−hour exposure cycles. Each 96−hour exposure cycle is defined as 48 hours of salt fog challenge followed by 48 hours of storage at ambient laboratory conditions with no salt fog challenge. After the five 96−hour exposure cycles, rinse the external surfaces with clear water if desired, dry, install/remove the filter in its housing, and inspect filter for evidence of damage and corrosion. Then test each gas filter for airflow resistance and leakage in accordance with 4.3.6.2 and 4.3.6.3, respectively, and test each particulate filter for filtration efficiency in accordance with MIL−PRF−51526.

4.3.6.10 Moisture content. Determine the moisture content of the adsorbent in accordance with ASTM D2867, Oven−Drying Test Method, except that the oven temperature shall be 103 to 107°C (217 to 224°F) and the drying time shall be 3 hours. The moisture content may also be determined through the use of moisture teller devices (e.g. halogen radiator) if approved by the Government (see 6.5). If the moisture content of the adsorbent exceeds the requirement, the filters represented by the samples shall be rejected.

5. PACKAGING

5.1 Packaging. For acquisition purposes, packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department or Defense Agency's automated packaging files, CD−ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The filter set, gas–particulate, 340 cmh (200 cfm) is intended for use as part of a Nuclear Biological Chemical (NBC) air handling system of a collective protection system. The Type I filter set will be used in the AICPS. The Type II filter set may be used in the shipboard CPS, in land–based CPE such as the Modular CPE and Simplified CPE, and in the transportable CPS.

6.2 Acquisition requirements.

(a) Title, number, and date of this specification.
(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced
(c) First article:
    (1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract when testing is performed by the Government.
    (2) Name and address of test facility and shipping instructions when testing is performed by the Government.
    (3) Time required for the Government to notify the contractor whether or not to proceed with production.
(d) Packaging requirements for the filter set.
(e) Additional identification and marking requirements such as manufacturer's name, serial number, date of manufacture, and contract number must be in the contract and cite the responsibility for the items affected.
(f) Whether the item of acquisition is a Type I filter set, a Type II filter set.
(g) Whether any Government Furnished Equipment is supplied.
(h) Whether publication number 0099 from the American Conference of Government Industrial Hygienists is the latest number for defining Threshold Limit Values

6.3 Design notes. Examples of gas filters are as follows:

Type I gas filter – Drawing 5-19-11704
Type II gas filter – Drawing 53711-6573660 (Navy), 5-19-6368 (Army)

6.4 Leakage detector. A halide leak detector manufactured by Nucon Systems, Inc. has been found suitable for this purpose. When using this detector, the presence of high vapor pressure halogenated contaminants in the refrigerant could interfere with the filter leak test resulting in false indications of filter leakage. Should false leakage indications be suspected, the refrigerant should be checked for the presence of such contaminants.

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6.5 Adsorbent media. Although manufacturers may consider using alternate adsorbent media, development tests comparable to or beyond those done on ASZM–TEDA carbon per EA–DTL–1704 would first have to be performed by the contractor and approved for use by the Government. Contractors are forewarned that such testing, which includes gas life capacity testing using a variety of chemical agents before and after open–air environmental exposure, may be lengthy and costly. All development and test costs would be borne by the contractor.

6.6 Subject term (key word) listing.

AICPS
Collective protection
CK
DMMP
Filtered air
Gas filter
MCPE
Particulate filter
SACP
SCPE

Custodian:

Army – EA

Preparing activity:

Army – EA

(Project No. 4240 – A252)

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE *(YYYYMMDD)*
	MIL-PRF-51527(EA)	20000313

3. DOCUMENT TITLE

Filter Set, Gas-Particulate, 340 CMH (200 CFM)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

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